Programming Serial Communication with I^2C

ECE 2564, Embedded Systems

Fall 2025

Sample Code

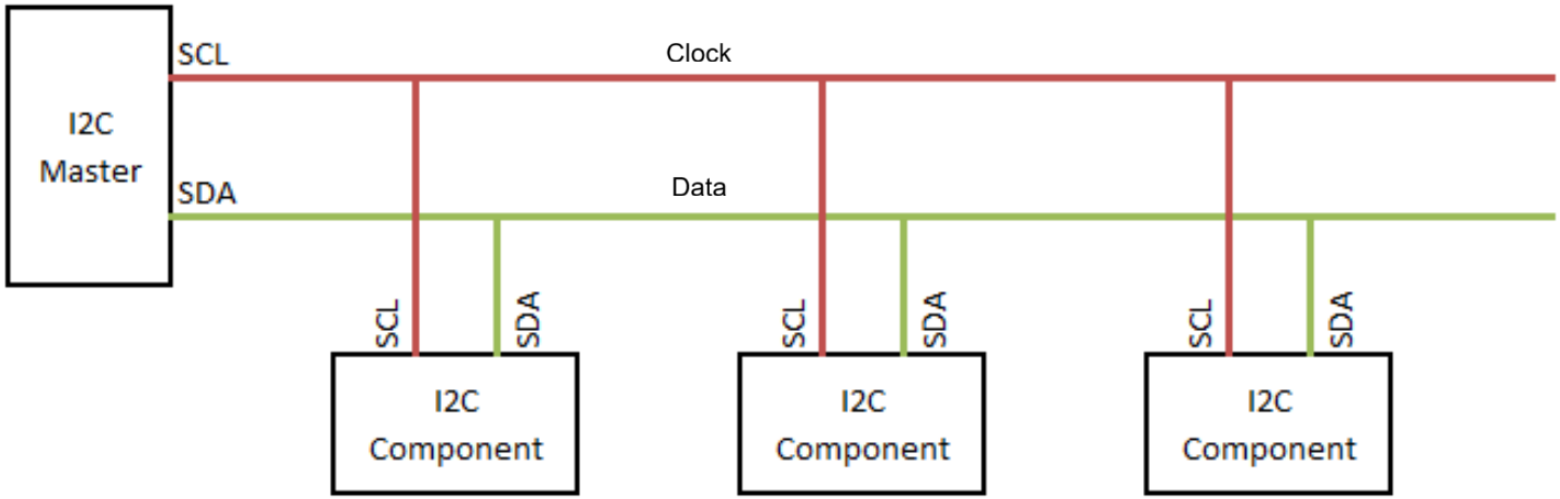
- You have installed the MSP432 SDK
- TI has provided us with an extensive collection of example projects
- The examples are at

```
C:\ti\simplelink_msp432p4_3_40_01_02\examples
```

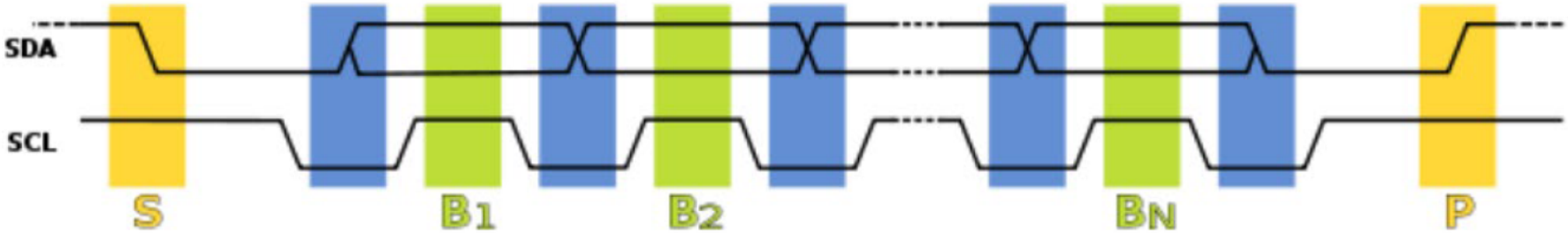
- Click **https://sites.google.com/vt.edu/introduction-to-embeddedsystem/tutorials/example-projects-from-ti** for instructions for importing the example projects into your CCS workspace

Reminder

- I^2C: Inter-Integrated Circuit
- I^2C is another serial communication standard
  - Different from using a UART
- SCL: one-wire serial clock
- SDA: one-wire serial data
- I^2C is serial, synchronous, half-duplex communication

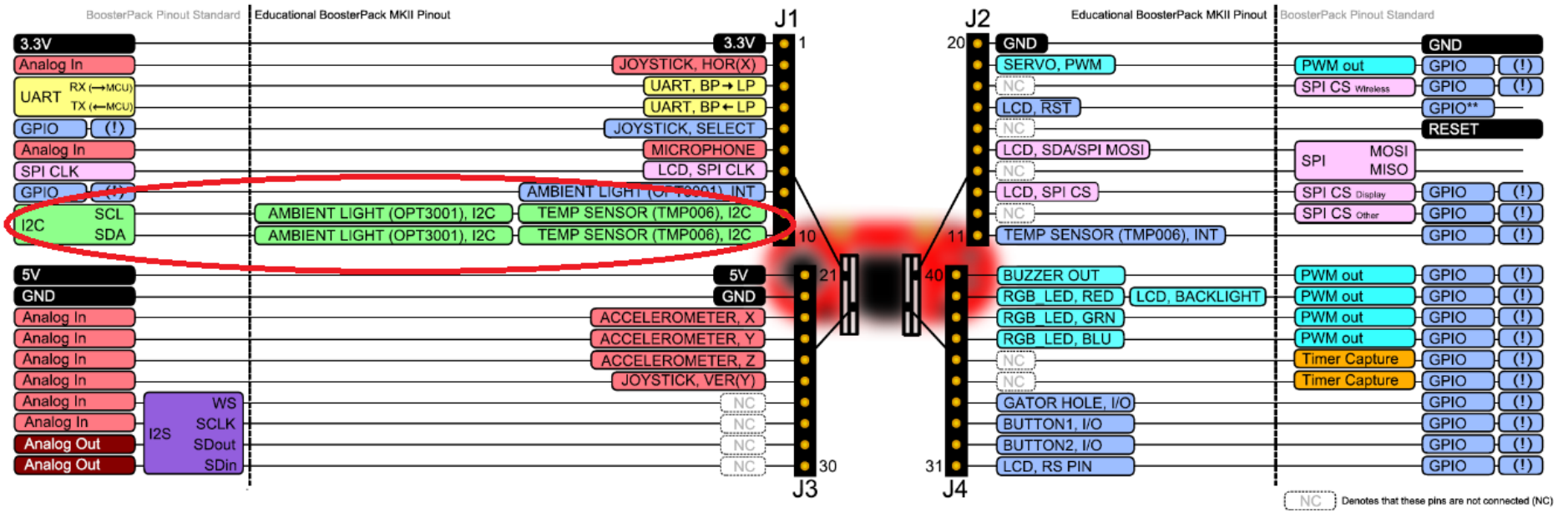


I^2C Communication Protocol - Recap



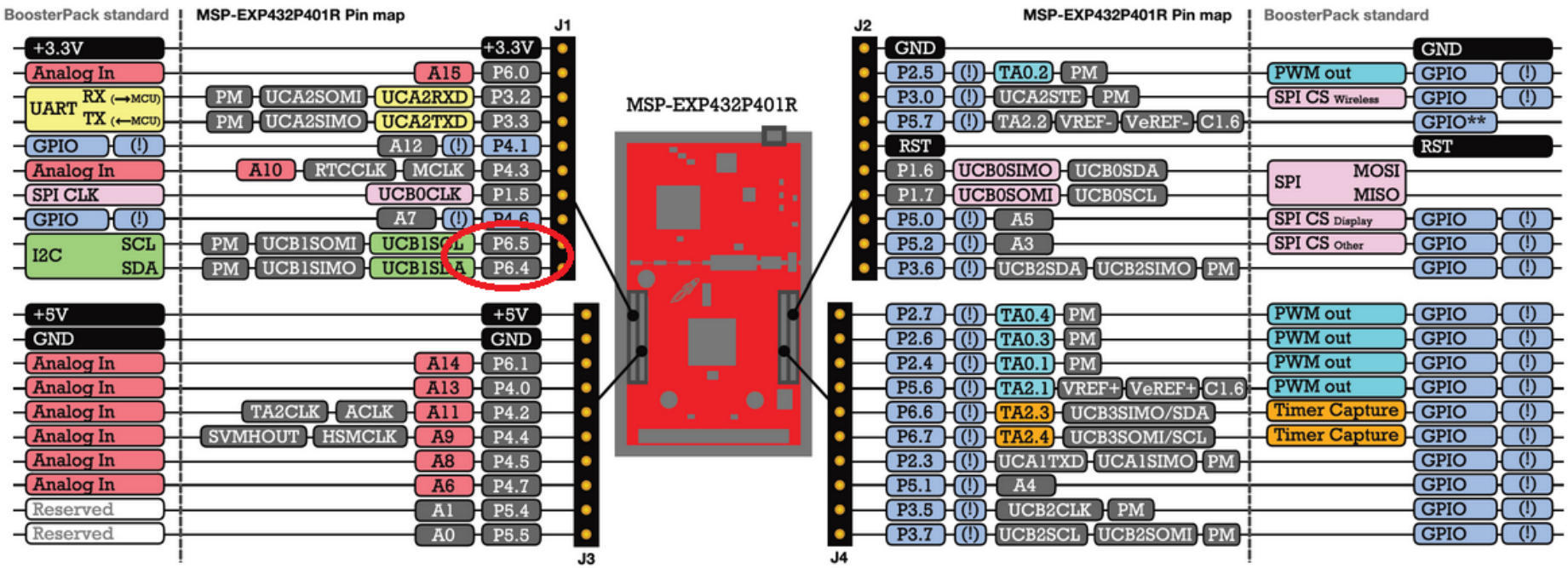
- SDA: serial data line
- SCL: serial clock
- S: start condition — pull SDA low while SCL is high
- B_i data bits — data is transmitted MSB *first* and is safe after the *rising (low-to-high transition) clock edge*
- P: stop condition — release SDA while SCL is high

What Devices use I^{2}C on our MSP432 BoosterPack?



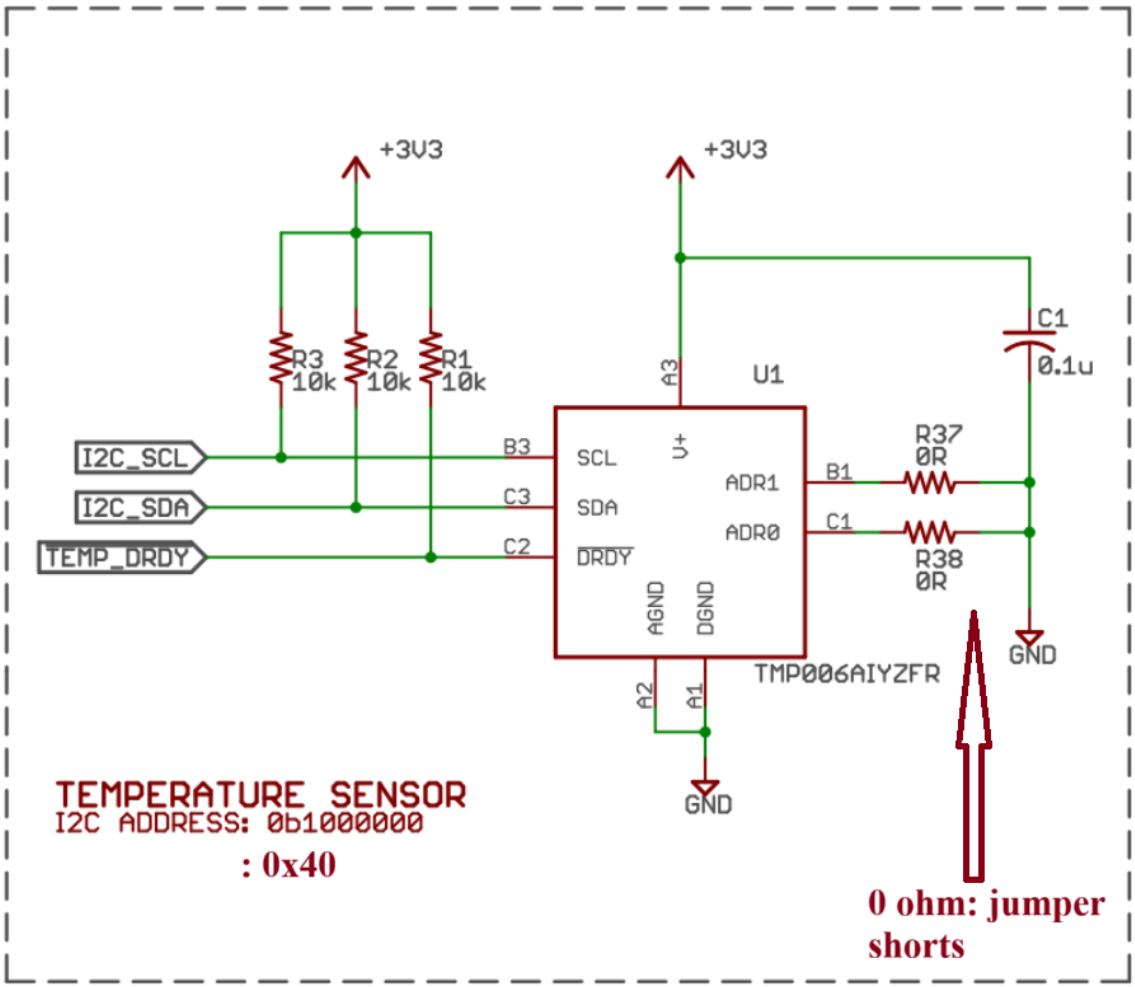
Temperature Sensor and Light Sensor

What Ports/Pins Correspond to the I^{2}C Devices?



Port/Pin: P6.4 and P6.5; UCB1

The Temperature Sensor on our MSP432 BoosterPack[1]



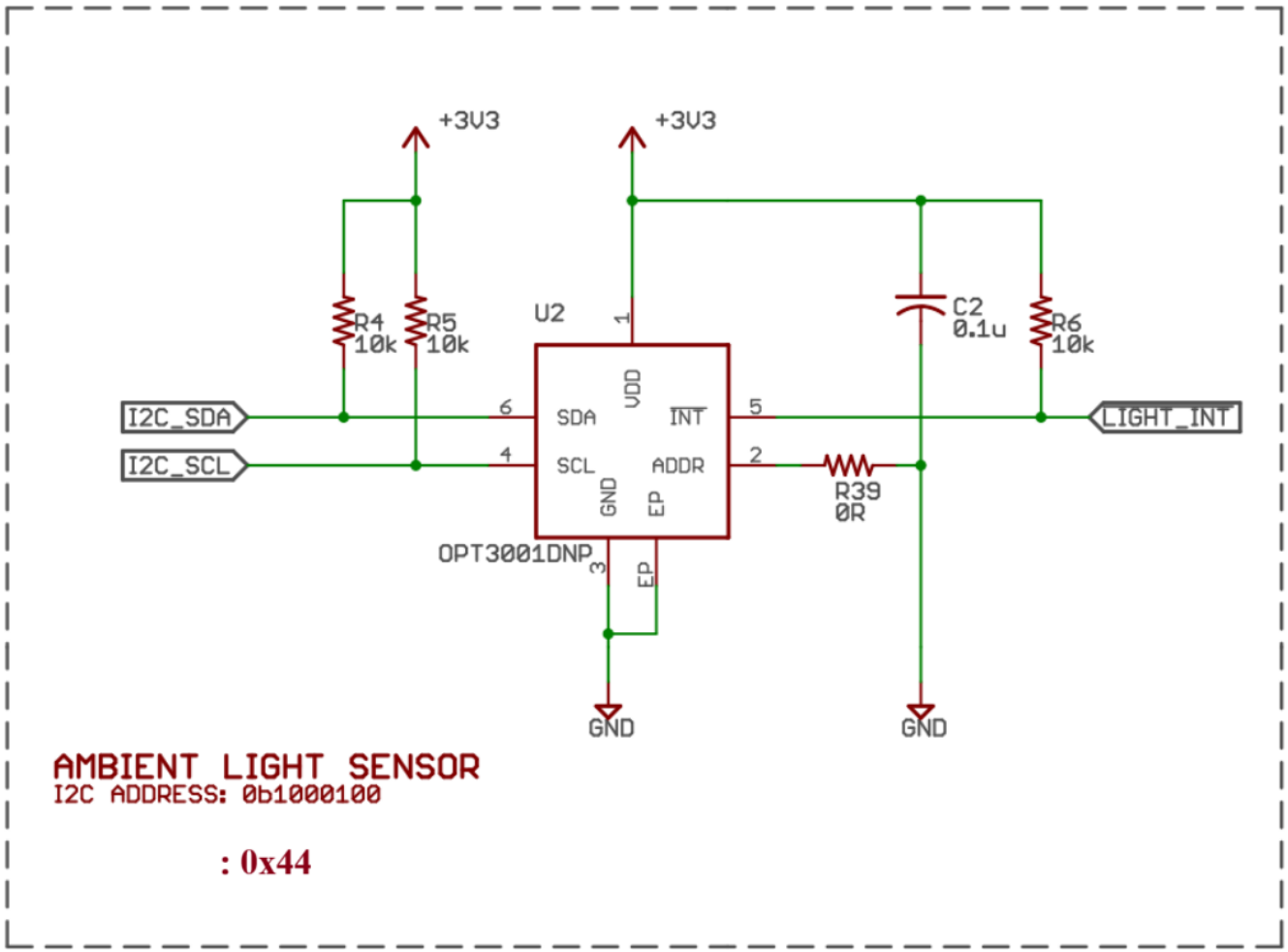
[1] From BoosterPack User Guide, Page 17

The I^{2}C Temperature Sensor Address

- The temperature sensor
  - P6, pins 4 and 5; UCB1
  - Can have up to eight slave temperature sensors
  - Configure ADR0/1 based on the table below to denote the corresponding slave address

ADR1	ADR0	SMBus Address
0	0	1000000
0	1	1000001
0	SDA	1000010
0	SCL	1000011
1	0	1000100
1	1	1000101
1	SDA	1000110
1	SCL	1000111

The Light Sensor on our MSP432 BoosterPack[2]



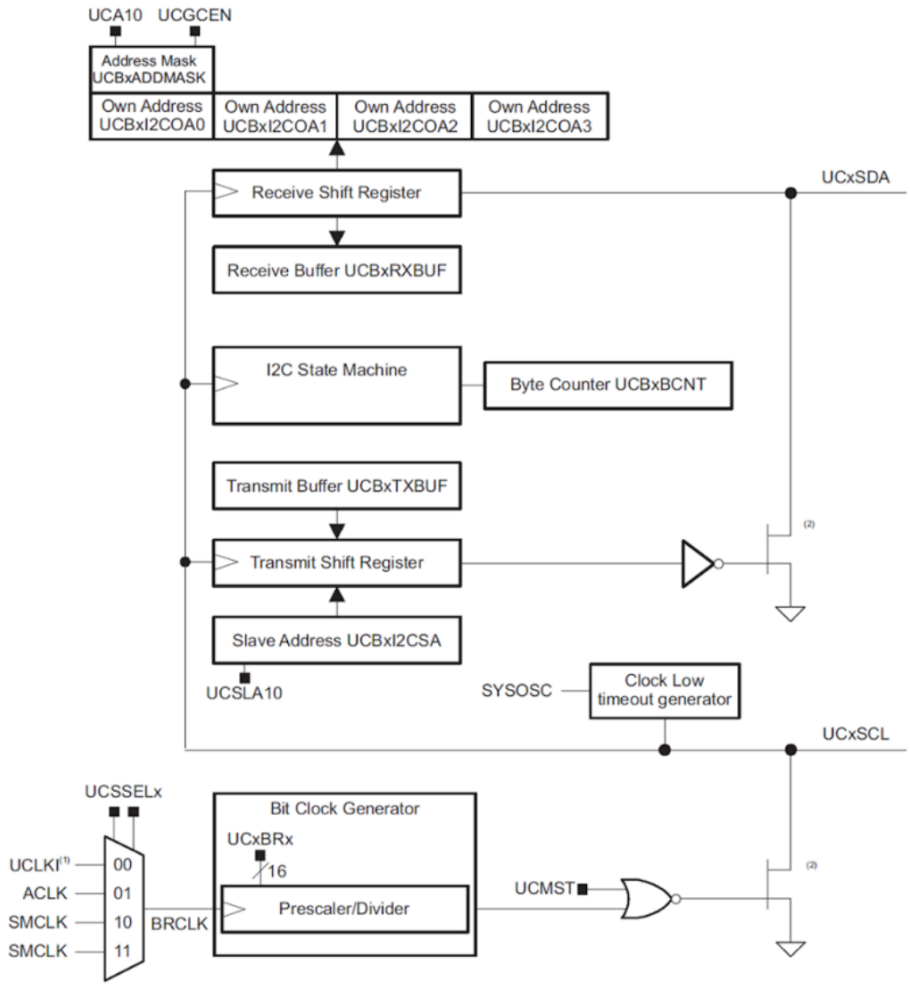
[2]From BoosterPack User Guide, Page 17

The I^{2}C Light Sensor Address

- The light sensor
  - Also P6, pins 4 and 5; UCB1
  - Can have up to four slave light sensors
  - Check OPT3001 guide for the configuration of the address

DEVICE I^{2}C ADDRESS	ADDR PIN
1000100	GND
1000101	VDD
1000110	SDA
1000111	SCL

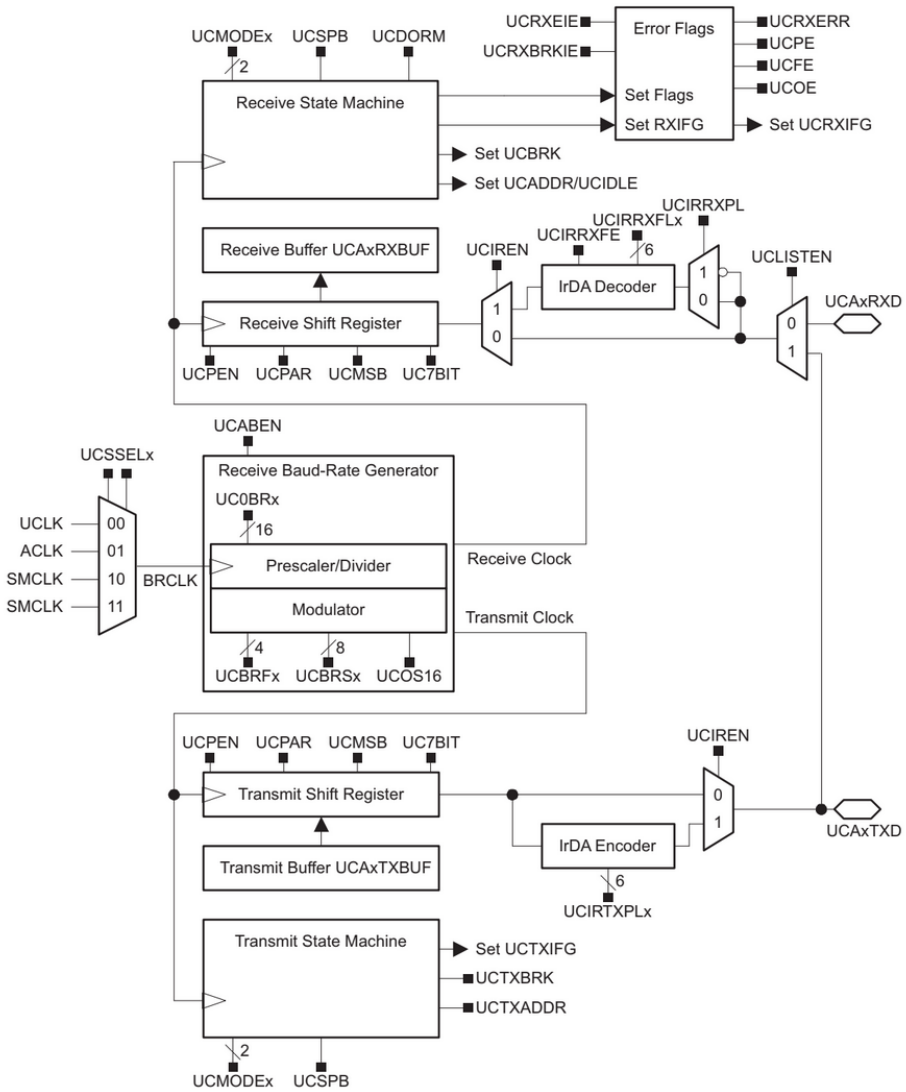
I²C Block Diagram (Figure 26-1 MCU Tech. Ref. Manual)



- Two open-drain transistors
  - SDA at top
  - SCL at bottom
- Why do we need shift registers?
  - Think *serial* transmission
- Why do we need the inverter
  - Think sending a “1” bit
- Half-duplex
  - Rx/Tx registers are tied to the same SDA signal

Recall UART (Figure 24-1 MCU Tech. Ref. Manual

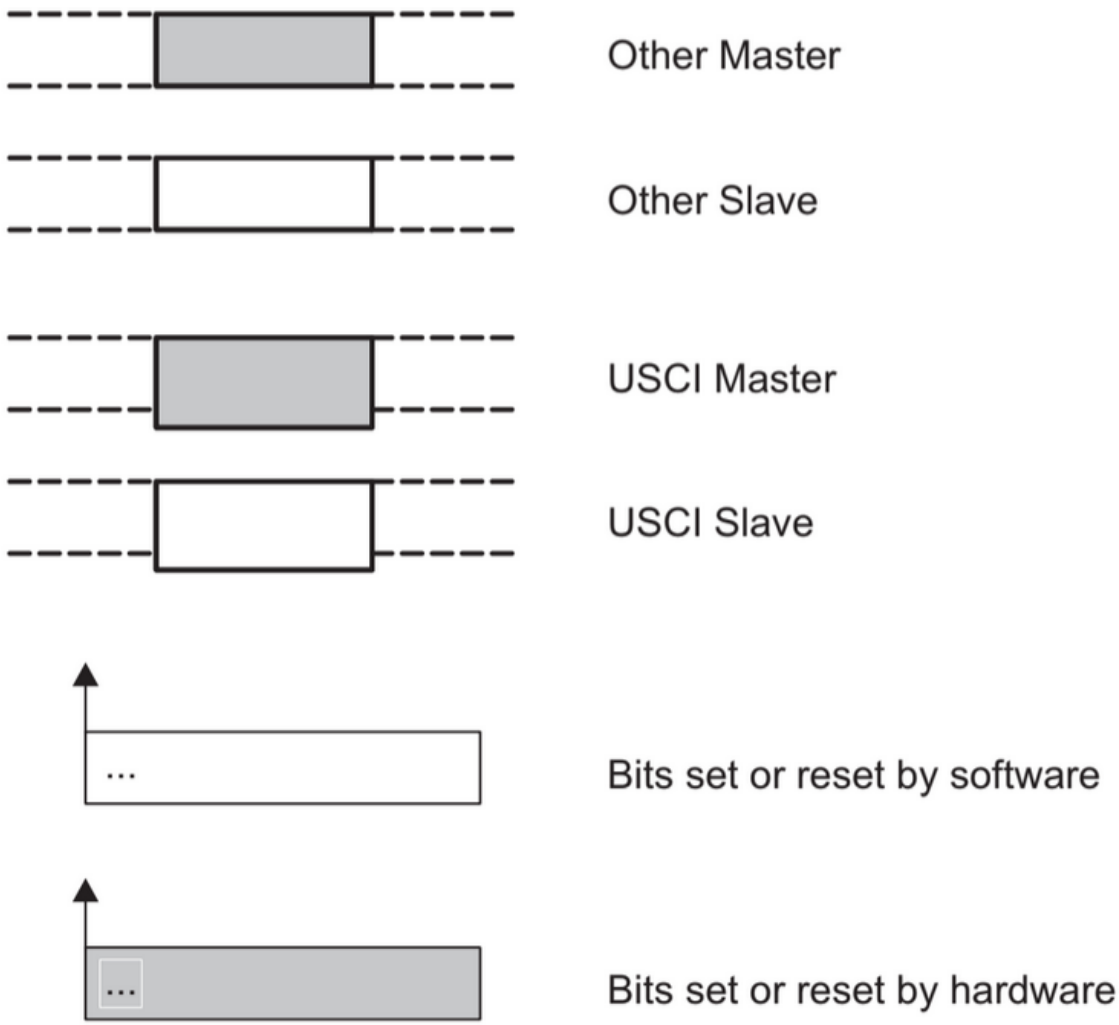
- Compare UART with I^2C
  - Full-duplex
  - No inverter (Why not?)
  - Two shift registers (Why?)
  - Clock divider



Four Configurations for the MSP432 I^2C device

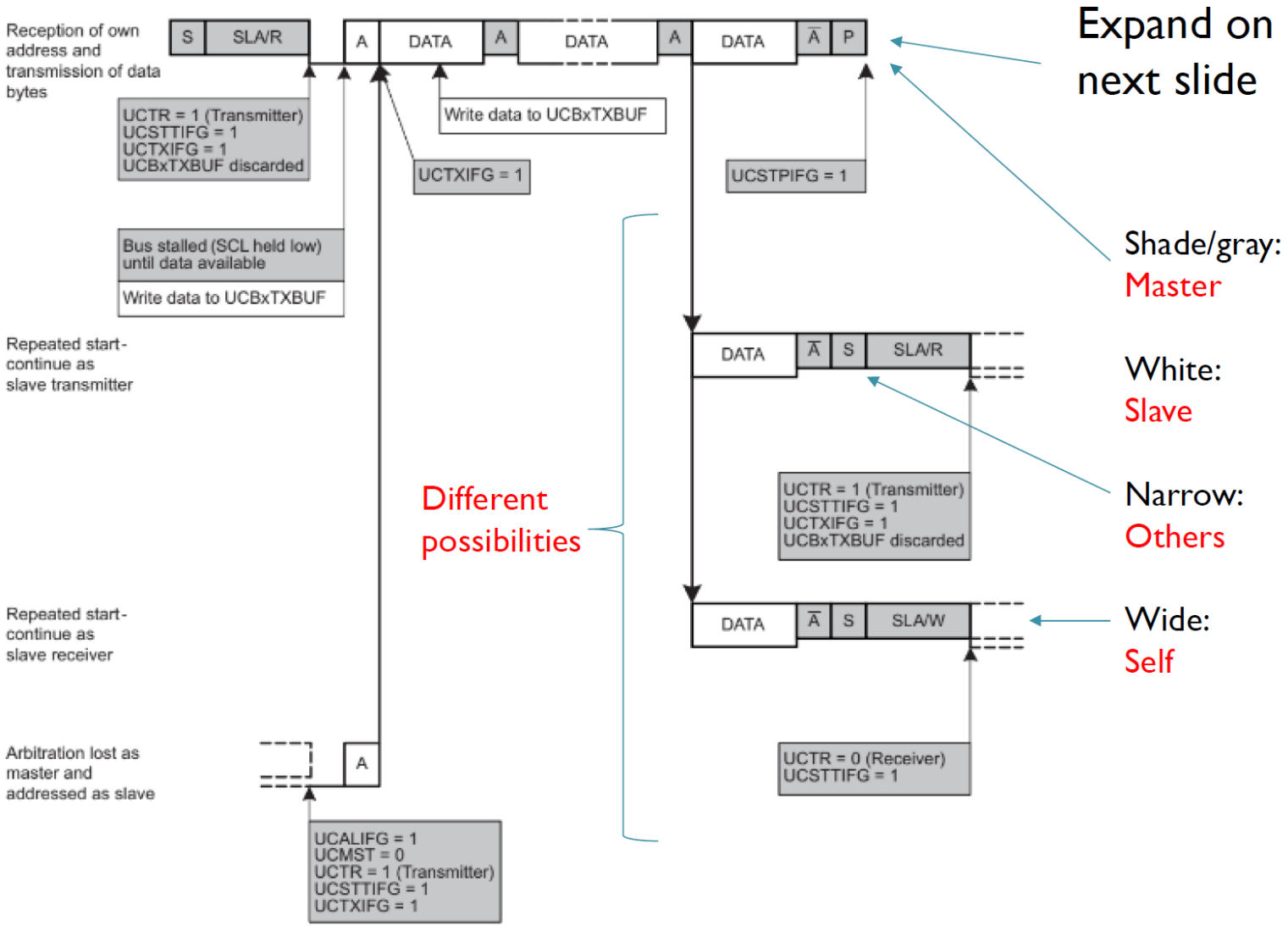
	Transmitter	Receiver
Master	Master Transmitter Mode	Master Receiver Mode
Slave	Slave Transmitter Mode	Slave Receiver Mode

I²C Timing Diagram Legend[3]



[3] Figure 26-8 in MCU Tech. Ref. Manual

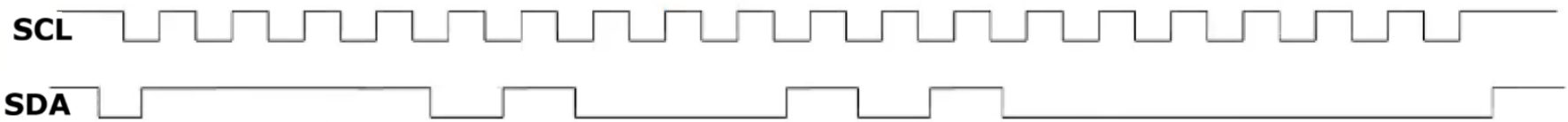
Timing Diagram for Master as Transmitter[4]



[4]

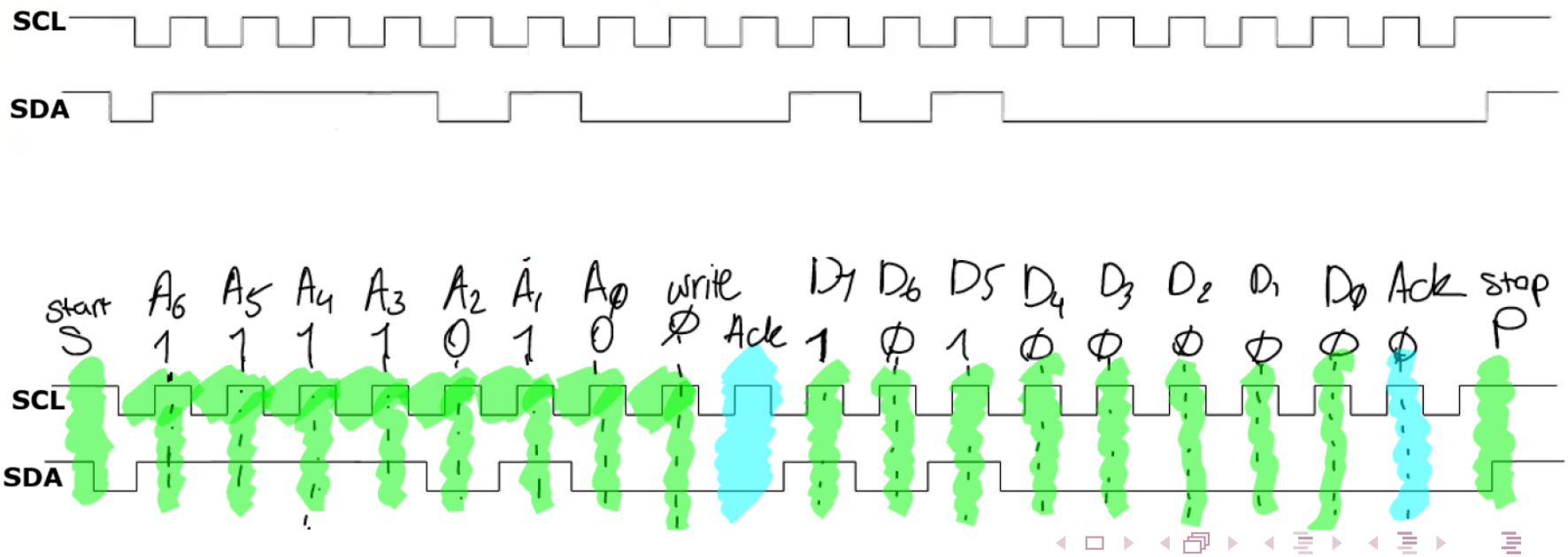
Decoding a Timing Diagram for Master as Transmitter

- In a timing diagram let's decide:
  - Who generates the start and stop condition?
  - What is the slave address?
  - Does the master want to read or write?
  - Who sends the first ACK (active-low)?
  - What is the data byte being sent?
  - Who sends the second ACK?



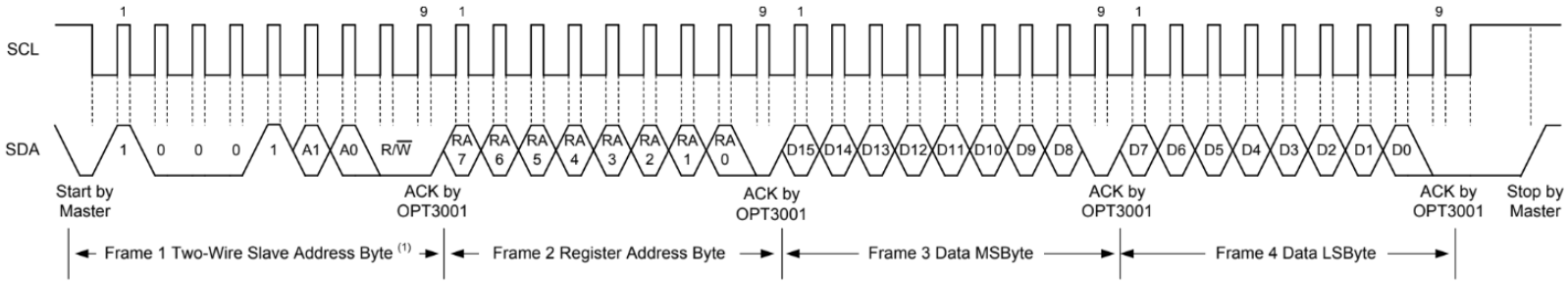
Decoding a Timing Diagram for Master as Transmitter

- In a timing diagram let's decide: Master-blue; Slave-green
  - Who generates the start and stop condition? Master
  - What is the slave address? 0x7A
  - Does the master want to read or write? Write
  - Who sends the first ACK (active-low)? Slave
  - What is the data byte being sent? 0xA0
  - Who sends the second ACK? Slave



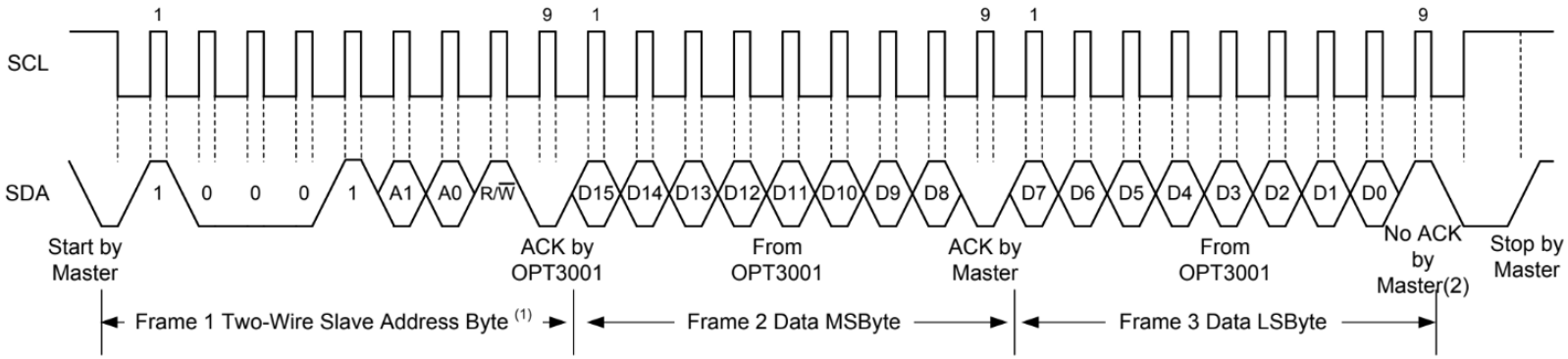
Timing Diagram for Ambient Light Sensor[5]

Figure 20 and Figure 21 show the write and read operation timing diagrams, respectively. Note that register bytes are sent most significant byte first, followed by the least significant byte.



(1) The value of the slave address byte is determined by the setting of the ADDR pin; see Table 1.

Figure 20. I²C Write Example



(1) The value of the slave address byte is determined by the ADDR pin setting; see Table 1.

(2) An ACK by the master can also be sent.

[5] Figures 20 and 21 from OPT3001 Manual

Summary

- I^2C Protocol
  - Serial, synchronous, half-duplex
  - SCL and SDA: both are open-drain connections
  - Protocol of transmission
  - Arbitration
- Example operations
  - Temperature sensor
  - OPT3001 ambient light sensor